



# Analysis of book circulation data and a book recommendation system in academic libraries using data mining techniques

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## ABSTRACT

The use of data mining modern technology in library management systems and information centers is of great importance. With the increasing availability of a large quantity of information, traditional tools and practices without wasting time and cost cannot respond to users accurately and quickly. The present study aims to analyze book circulation transactions and discover the user's book loan patterns to develop a recommender system. The data included 109,639 transactions and information from 8636 user records. Microsoft SQL Server and Matlab software were applied to analyze the data. Item-based collaborative filtering algorithms and decision tree methods were also applied. The results led to the extraction of rules for suggesting books to users. Analysis of the circulation data could be applied to address many issues like evaluation, collection acquisition policies, allocating funding for materials, and suggesting approaches to deselecting and allocating physical space for materials.

## 1. Introduction

Data mining has become the most active research database technique, and many scientists have studied its applications and theories, and the results of their investigations have provided a proper theoretical framework for applying data mining in academic libraries (Huang, Li, & Xiao, 2018; Liu, 2018; Ochilbek, 2019; Philip, Haugen, Lener, Pan-nabecker, & Brittle, 2017; Rattan, 2019). Applying data mining techniques in academic libraries leads to discovering hidden patterns of users' information behaviors and ultimately leads to a collection development based on users' real needs for information (Krishnamurthy & Balasubramani, 2014; Siguenza-Guzman, Saquicela, Avila-Ordóñez, Vandewalle, & Cattrysse, 2015). In order to analyze users' information searching behaviors and be aware of their information needs, academic libraries must use new methods and technologies; data mining techniques are among the most important of these approaches. Therefore, the application and development of data mining techniques are essential for the databases of academic libraries.

Academic libraries use various information technologies to manage big data, such as loan data, and can only succeed in satisfying users' needs if they can anticipate the potential information needs of users and

provide suitable solutions for meeting them. For this purpose, data related to users' previous activities should be investigated for necessary analyses and predictions. Therefore, it seems essential to predict the usage rate of users and the amount of their need for available resources (number of loans). In this regard, the loan circulation of information resources provides researchers with valuable information for data mining and improvement of the circulation system of academic libraries (Yi, Chen, & Cong, 2018). Resource circulation analysis is one of the conventional approaches for evaluating the academic library collection (Philip et al., 2017). The results of this analysis can improve the quality of services, provide latent knowledge of data, discover meaningful relationships and patterns between book-loan data, users' data, and library resources, and identify and analyze frequently used resources (Yu, 2011). The results can also manage financial resources, budgeting, and strategic decisions (Rattan, 2019). To make better decisions, interact effectively with users, and suggest related information resources, the circulation department needs to know users' usage rates, search patterns, library resources, and academic fields. (Jomsri, 2014; Liu, 2018; Silwattananusarn & Kulkanjanapiban, 2020; Uppal & Chindwani, 2013).

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